

Open to opportunities

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Most of these projects were built with AI-assisted development: I direct the architecture and requirements, use LLM tooling for implementation, and review, test, and verify the results myself.

AI, LLMs & Applied GenAI

modelica-genai-lab github.com/gericapo98/modelica-genai-lab

Automated Modelica model generation, compilation & verification with GenAI. Pipeline: prompt → .mo model → OpenModelica simulation → numeric comparison (RMSE/tolerance) → optional STL property checks → deviation report.

GenAI · Modelica · OMPython · FMU/FMPy · RTAMT (STL) · Verification

langchain-rpa-simkit github.com/0x000-byte/langchain-rpa-simkit

LangChain toolkit that simulates RPA / agent workflows with typed (pydantic) chains, packaged with uv and covered by tests.

LangChain · Agents · Python · pydantic

intent-engine github.com/0x000-byte/intent-engine

Natural-language intent engine: candidate generation, NL parsing, and ranking to map free text onto structured intents.

NLP · Intent detection · Python

llms-lab github.com/gericapo98/llms-lab

Experiments in LLM reasoning — DeepSeek-Math evaluation and rStar self-play reasoning tests.

LLMs · Reasoning · Evaluation

asr local project

Whisper-based pipeline that transcribes lecture audio into aligned SRT / VTT / TSV transcripts.

ASR · Whisper · Python

Robotics & Autonomy

ros2-common-setup github.com/0x000-byte/ros2-common-setup

Reusable ROS 2 Rolling environment setup, including bring-up on macOS.

ROS 2 · DevOps · macOS

robot_sim local project

A ROS 2 simulation playground for trying out robotics ideas.

ROS 2 · Simulation

Data Engineering & Visualization

pyduck-iceberg github.com/gericapo98/pyduck-iceberg

A DuckDB + Apache Iceberg lakehouse pipeline with a command-line interface and a fully Dockerized stack.

DuckDB · Apache Iceberg · Lakehouse · CLI · Docker

visualization-lab github.com/0x000-byte/visualization-lab

Hybrid toolkit for dynamic animations with Manim (Python) and plots with Julia — built for explainers, demo videos, and interactive storytelling.

Manim · Julia · Animation · SVG

Systems, Security & Tooling

mangodex-client github.com/gericap98/mangadex-go-cli

A Go command-line client for the MangaDex API, with OAuth token auth and OS-keyring refresh-token storage.

Go · CLI · REST API

system-design-templates [local project](#)

A single-server system-design reference: a Node/Express web service with a SQL schema and JWT/bcrypt auth.

Node.js · Express · SQL · JWT

ubuntu-sandbox github.com/0x000-byte/ubuntu-sandbox

A reproducible Ubuntu development sandbox provisioned via cloud-init.

Docker · Linux · IaC · cloud-init

kahoot-dl [local project](#)

Python tool to bulk-download all public Kahoot quizzes for a given creator.

Python · CLI · Scraping

cv-portfolio [local project](#)

A self-hosted single-page CV/portfolio site of static files served by nginx in Docker — the source of this very document.

HTML · JavaScript · nginx · Docker

Foundations & Explorations

machine-learning [local project](#)

ML coursework and notebooks covering clustering, expectation-maximization, and supporting notes.

ML · Clustering · EM · Notebooks

resyst-study [local project](#)

A self-built web study app (quiz + materials) for a Reactive Systems course.

Web · JavaScript · Study tool

Open-Source Ecosystems — Studied & Worked With

Listed for transparency: third-party codebases I have studied, run, or contributed to — not authored by me.

ServePRO — tennis/badminton serve analyzer (audio detection + CV trajectory fitting); *contributor*.

Hugging Face Transformers — studied/used for LLM workflows.

Ollama — ran the local LLM runtime.

Apple MLX (mlx-lm) — running and fine-tuning LLMs on Apple silicon.

Odysseus — local-first self-hosted AI workspace; deployed and operate my own instance.

arena2d — deep-RL robot navigation in 2D simulation.

flying_adversarial_patch — adversarial patches against multirotor pose estimation.

Snort 3 — network intrusion detection & prevention system.

Zeek — network-security monitoring.